

DS MINI PROJECT ACTIVITY

Title :- Vehicle Parking Management System

Group Members :-

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Div :- C Batch :- C4

Code :

```
#include <iostream>
using namespace std;

// ----- STACK (EXIT HISTORY) -----
class Stack
{
    int top;
    string arr[50];

public:
    Stack()
    {
        top = -1;
    }

    void push(string x)
    {
        if (top == 49)
        {
            cout << "Stack Full\n";
        }
        else
        {
            top++;
            arr[top] = x;
        }
    }
}
```

```

void displayAll()
{
    if (top == -1)
    {
        cout << "No Exit History\n";
    }
    else
    {
        cout << "Exit History: ";
        for (int i = top; i >= 0; i--)
        {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
}
};

```

// ----- QUEUE (ENTRY SYSTEM) -----

```

class Queue
{
    int front, rear;
    string arr[50];

public:
    Queue()
    {
        front = -1;
        rear = -1;
    }

    void enqueue(string x)
    {
        if (rear == 49)
        {
            cout << "Queue Full\n";
            return;
        }

        if (front == -1)
            front = 0;

        rear++;
    }
}

```

```

        arr[rear] = x;
    }

void display()
{
    if (front == -1)
    {
        cout << "Queue Empty\n";
        return;
    }

    cout << "Entry Queue: ";
    for (int i = front; i <= rear; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}
};

// ----- PARKING SYSTEM -----
class Parking
{
    string vehicle[3][5]; // ARRAY (MAIN)

    Queue q; // Queue
    Stack s; // Stack

public:
    Parking()
    {
        for(int i=0;i<3;i++)
        {
            for(int j=0;j<5;j++)
            {
                vehicle[i][j] = "EMPTY";
            }
        }
    }

    // DISPLAY
    void display()
    {
        cout<<"\n--- Parking Status ---\n";
    }
};

```

```

for(int i=0;i<3;i++)
{
    cout<<"Floor "<<i+1<<": ";
    for(int j=0;j<5;j++)
    {
        if(vehicle[i][j]=="EMPTY")
            cout<<"[Free] ";
        else
            cout<<"[Booked] ";
    }
    cout<<endl;
}
}

// BOOK
void book()
{
    string num;
    int f,s1;

    cout<<"Enter Vehicle Number: ";
    cin>>num;

    // Duplicate check
    for(int i=0;i<3;i++)
    {
        for(int j=0;j<5;j++)
        {
            if(vehicle[i][j]==num)
            {
                cout<<"Vehicle Already Parked\n";
                return;
            }
        }
    }
}

cout<<"Enter Floor (1-3): ";
cin>>f;
cout<<"Enter Slot (1-5): ";
cin>>s1;

if(vehicle[f-1][s1-1]=="EMPTY")
{

```

```

        vehicle[f-1][s1-1]=num;

        q.enqueue(num); // Queue use

        cout<<"Booked Successfully\n";

        int h;
        cout<<"Enter hours: ";
        cin>>h;

        cout<<"Payment = "<<h*20<<" Rs\n";
    }
    else
    {
        cout<<"Slot Already Booked\n";
    }
}

// EXIT
void exitVehicle()
{
    string num;
    cout<<"Enter Vehicle Number: ";
    cin>>num;

    for(int i=0;i<3;i++)
    {
        for(int j=0;j<5;j++)
        {
            if(vehicle[i][j]==num)
            {
                vehicle[i][j]="EMPTY";

                s.push(num); // Stack use

                cout<<"Vehicle Exited\n";
                return;
            }
        }
    }

    cout<<"Vehicle Not Found\n";
}

```

```

// SEARCH
void search()
{
    string num;
    cout<<"Enter Vehicle Number: ";
    cin>>num;

    for(int i=0;i<3;i++)
    {
        for(int j=0;j<5;j++)
        {
            if(vehicle[i][j]==num)
            {
                cout<<"Found at Floor "<<i+1<<" Slot "<<j+1<<endl;
                return;
            }
        }
    }

    cout<<"Not Found\n";
}

// SHOW STACK
void showExitHistory()
{
    s.displayAll();
}

// SHOW QUEUE
void showQueue()
{
    q.display();
}
};

// ----- MAIN -----
int main()
{
    cout<<"=====\n";
    cout<<"  Parking Management System\n";
    cout<<"=====\n";

    cout<<"\nTeam Members:\n";
    cout<<"1) Nakul Nitesh Lachure (PRN: 202502070024)\n";
}

```

```

cout<<"2) Om Gajanan Sapdhare (PRN: 202502070030)\n";

cout<<"\nLocation: Phoenix Mall Pune\n";
cout<<"=====\n";

Parking p;
int ch;

do
{
    cout<<"\n1.Display\n2.Book\n3.Exit\n4.Search\n5.Exit History\n6.Entry
Queue\n7.Exit\n";
    cout<<"Enter choice: ";
    cin>>ch;

    switch(ch)
    {
        case 1: p.display(); break;
        case 2: p.book(); break;
        case 3: p.exitVehicle(); break;
        case 4: p.search(); break;
        case 5: p.showExitHistory(); break;
        case 6: p.showQueue(); break;
        case 7: cout<<"Exit\n"; break;
        default: cout<<"Invalid\n";
    }

}while(ch!=7);

return 0;
}

```

Output :

1) Initial Parking Status

```
=====
      Parking Management System
=====

Team Members:
1) Nakul Nitesh Lachure (PRN: 202502070024)
2) Om Gajanan Sapdhare (PRN: 202502070030)

Location: Phoenix Mall Pune
=====

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 1

--- Parking Status ---
Floor 1: [Free] [Free] [Free] [Free] [Free]
Floor 2: [Free] [Free] [Free] [Free] [Free]
Floor 3: [Free] [Free] [Free] [Free] [Free]
```

2) Booking Status

```
Enter choice: 2
Enter Vehicle Number: MH129420
Enter Floor (1-3): 1
Enter Slot (1-5): 3
Booked Successfully
Enter hours: 3
Payment = 60 Rs

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 2
Enter Vehicle Number: MH124203
Enter Floor (1-3): 2
Enter Slot (1-5): 4
Booked Successfully
Enter hours: 5
Payment = 100 Rs

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 1

--- Parking Status ---
Floor 1: [Free] [Booked] [Booked] [Free] [Free]
Floor 2: [Free] [Free] [Free] [Booked] [Free]
Floor 3: [Free] [Free] [Free] [Free] [Free]
```


3) Exit Status

```
1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 3
Enter Vehicle Number: MH124203
Vehicle Exited

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 3
Enter Vehicle Number: MH122430
Vehicle Exited

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 1

--- Parking Status ---
Floor 1: [Free] [Free] [Booked] [Free] [Free]
Floor 2: [Free] [Free] [Free] [Free] [Free]
Floor 3: [Free] [Free] [Free] [Free] [Free]
```

4) Searching Vehicle

```
1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 2
Enter Vehicle Number: MH124574
Enter Floor (1-3): 1
Enter Slot (1-5): 4
Booked Successfully
Enter hours: 5
Payment = 100 Rs

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 4
Enter Vehicle Number: MH124574
Found at Floor 1 Slot 4
```

5) Exit History

```
1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 3
Enter Vehicle Number: MH124574
Vehicle Exited

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 5
Exit History: MH124574 MH122430 MH124203
```

6) Entry History

```
1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 5
Exit History: MH124574 MH122430 MH124203

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 6
Entry Queue: MH122430 MH129420 MH124203 MH124574
```

7) System Exit

```
1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 5
Exit History: MH124574 MH122430 MH124203

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 6
Entry Queue: MH122430 MH129420 MH124203 MH124574

1.Display
2.Book
3.Exit
4.Search
5.Exit History
6.Entry Queue
7.Exit
Enter choice: 7
Exit

-----
Process exited after 1808 seconds with return value 0
Press any key to continue . . .
```

CONCLUSION :-

The **Parking Management System** successfully demonstrates the use of basic data structures to solve a real-world problem of managing parking in a multi-level environment. The system allows efficient booking, tracking, relocation, and exit of vehicles using simple array-based implementation.

Key features such as **multi-floor management, vehicle number tracking, duplicate prevention, and overflow handling** make the system practical and realistic. The menu-driven approach ensures easy interaction, while real-time updates reflect the current parking status accurately.

This project highlights how data structures can be applied in real-life applications like mall parking systems. It also improves understanding of concepts like searching, updating, and managing data efficiently.

Overall, the project is simple, effective, and demonstrates a clear connection between theoretical concepts and practical implementation.